Hybrid Quantum AI for Accelerating Multiphysics Simulation

A Hybrid-QPINN / QPINO framework: Fourier features, data re-uploading and shallow parameterised quantum circuits

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Outline

Accurate multiphysics simulation is too slow for design loops

- High-fidelity coupled CFD (flow + heat + rotation + phase change) takes hours to days per run.
- The real cost is the **parameter sweep**: every operating point is a new simulation.
- Process control and digital twins need full fields in **milliseconds**.
- Classical surrogates (PINN, PINO) help, but hit training bottlenecks on sharp, strongly coupled fields.

[IMAGE PLACEHOLDER]
`images/cz-process-schematic.png`
Czochralski furnace schematic: crucible, melt, rotating crystal, heater

Goal

A surrogate that keeps the governing physics, generalises across operating conditions, and evaluates in real time.

Two industrial use cases

Silicon crystal growth (Czochralski)

- Seed crystal pulled from molten Si in a rotating crucible, 1685–1750 K.
- Melt convection, swirl and thermal structure set defect density and oxygen uptake.
- **Aim:** fast field prediction to reduce defect formation.

[IMAGE PLACEHOLDER]
`images/cz-melt-fields.png`
CZ melt temperature / velocity field

LH_2 cryogenic boil-off

- Heat leak into liquid-hydrogen tanks drives natural convection, stratification and evaporation.
- Turbulent convection with phase change and a moving liquid–vapour interface.
- **Aim:** predict boil-off rate and pressure rise for tank design.

[IMAGE PLACEHOLDER]
`images/lh2-tank-schematic.png`
LH_2 tank: heat leak, stratified layer, ullage vapour

Our proposal in one slide

1. **Problem:** classical PINNs suffer from spectral bias, loss imbalance and badly scaled coupled fields.
2. **Framework:** Hybrid-QPINN / QPINO, a classical network with a shallow parameterised quantum circuit (PQC) as a trainable feature map, fed by Fourier features and data re-uploading.
3. **Target physics:** strongly coupled non-linear transport: velocity, temperature, and (next) concentration and evolving interfaces.
4. **Benchmarks:** silicon CZ growth (running) and LH_2 boil-off (defined).
5. **Destination:** real-time digital twins and fast parametric exploration.

How to read the results

Each claim in this talk carries a status: **Shown** (measured), **In progress** (experiment running or partially done), or **Planned**.

Physics-informed neural networks in one slide

- A network $\hat{u}_\theta(x)$ is trained so that the PDEs and boundary/initial conditions hold.
- Derivatives come from automatic differentiation: no mesh, no stencil.

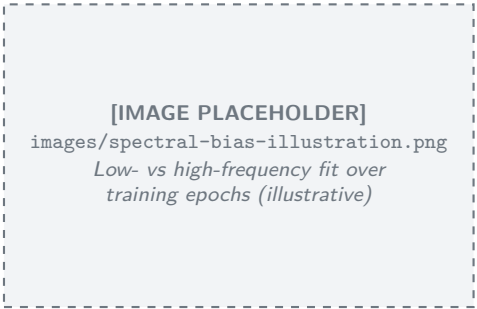
$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{PDE}}\,\mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}}\,\mathcal{L}_{\text{BC}} + \lambda_{\text{IC}}\,\mathcal{L}_{\text{IC}}$$

- Strength: little labelled data, natural fit for coupled equations.
- Limitation: one PINN = one parameter setting. Neural operators (PINO) learn whole solution families.



Bottleneck 1: spectral bias

- MLPs learn low frequencies first; high-frequency content converges slowly (Rahaman et al., 2019).
- Coupled transport fields are full of sharp structure: thermal boundary layers, wall shear, swirl near rotating walls, phase interfaces.
- Classical remedy: Fourier feature mapping of the inputs (Tancik et al., 2020).
- **Our addition:** a second, *trainable* non-linear feature map, the PQC, whose outputs are trigonometric polynomials of its inputs.



Bottleneck 2: loss-term imbalance

- The loss term with the largest raw gradient dominates the update (Wang et al., 2021).
- A small *total* loss can hide poor boundary satisfaction and a wrong field.

Trainer (1D heat equation)	PDE loss	BC/IC loss	Rel. L_2 error
Fixed weights	3.96×10^{-3}	1.15×10^{-3}	0.084
Normalised gradients	9.74×10^{-3}	9.05×10^{-4}	**0.054**
Adaptive weights	$\mathbf{8.22 \times 10^{-7}}$	2.00×10^{-1}	1.368

Lesson carried into this work

The smallest PDE residual gave the worst solution. We report loss components and per-field errors, never only a weighted total.

Chapagain et al., NeurIPS 2026 Sim2Science workshop.

Bottleneck 3: scale mismatch between coupled fields

- Five coupled outputs: u_r, u_z, u_θ, p, T, with very different magnitudes.
- In the CZ model the pressure coefficient is

$$p_{\text{coeff}} = \frac{p_{\text{scale}}}{v_{\text{scale}}^2} = 134{,}128.$$

- Viscous terms scale as $1/Re \approx 0.067$.
- Pressure-to-viscous ratio $\approx 2 \times 10^6 : 1$.

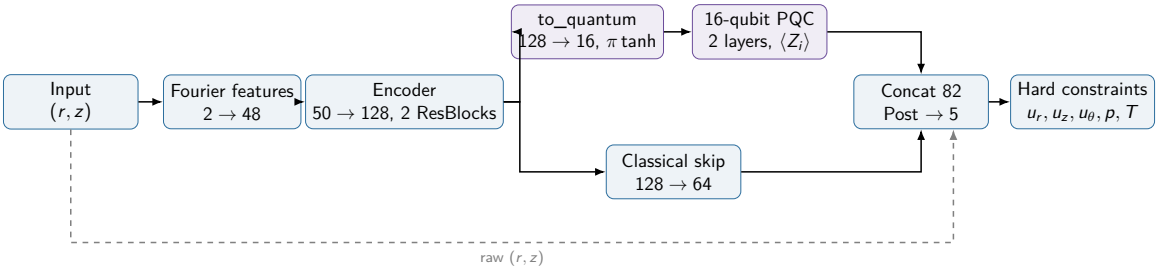
Consequence

The pressure residual can overwhelm the data signal during physics-informed training. This is the main stability issue we measured (see Results).

Design response

Hard constraints, staged physics warm-up, EMA residual balancing, and (next) per-field residual scaling.

Architecture overview



Classical backbone

Fourier encoding plus residual MLP: carries most of the computation.

Quantum feature map

16 qubits, 2 layers, 96 trainable angles; 16 expectation values out.

Skip path

Guarantees the head never depends solely on the quantum bottleneck.

Ingredient 1: Fourier feature mapping

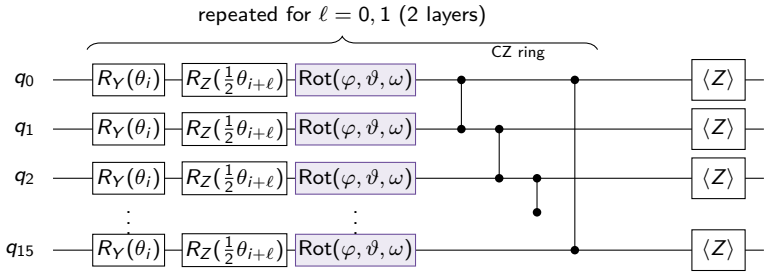
- Frozen random matrix $B \in \mathbb{R}^{2\times 24}$, $B_{ij} \sim \mathcal{N}(0, 6^2)$.

$$\gamma(x) = \left[\sin(2\pi\, xB),\ \cos(2\pi\, xB)\right] \in \mathbb{R}^{48}$$

- Frequencies span roughly ± 15 cycles per unit length in normalised coordinates.
- Concatenated with raw (r, z): 50 inputs to the encoder.
- Gives the network high-frequency access from the first layer, at zero trainable cost.

[IMAGE PLACEHOLDER]
`images/fourier-B-matrix.png`
Fourier feature frequencies, B matrix (2 × 24) heatmap

Ingredient 2: shallow parameterised quantum circuit



1. Encode: $R_Y(\theta_i)$, $\theta_i = \pi \tanh(h_i) \in [-\pi, \pi]$.
2. Re-upload: $R_Z(0.5\,\theta_{(i+\ell) \bmod 16})$.
3. Train: $\mathrm{Rot}(\varphi_i, \vartheta_i, \omega_i)$, 48 angles/layer.
4. Entangle: CZ ring, closing q_{15}–q_0.

Readout

$\langle Z_i \rangle$, $i = 0, \ldots, 15 \rightarrow$ LayerNorm(16).
96 trainable quantum parameters, about 50 gates per shot on hardware.

Ingredient 3: data re-uploading

- A circuit that encodes its input once realises only a low-degree Fourier series of that input.
- Re-uploading the data in every layer raises the accessible frequency spectrum (Pérez-Salinas et al., 2020; Schuld et al., 2021).
- The cyclic shift $\theta_{(i+\ell) \bmod 16}$ lets each layer mix a different combination of latent features.
- Together with Fourier inputs, this targets the high-frequency content that spectral bias suppresses.

Design rule

The PQC is a compact, expressive *feature extractor*. The classical network carries the bulk of the computation, which keeps the circuit shallow enough for today's hardware.

Parameter budget: a deliberately small quantum layer

Component	Parameters
Coordinate projector	23,040
ResidualMLP blocks (2×2)	66,048
to_quantum	9,296
Classical skip	8,256
Quantum circuit	**96**
Quantum LayerNorm	32
Post network	19,205
Fourier B (frozen)	48
Total	**126,021**

Why small?

- Shallow circuits survive NISQ noise.
- Few angles $\Rightarrow$ cheap hardware fine-tuning.
- 0.076% of all parameters.

Consequence for evaluation

Any gain must be attributed to the PQC by a controlled ablation, not assumed.

Governing equations (1/2): mass and momentum

Steady, axisymmetric, incompressible Navier–Stokes with Boussinesq buoyancy (nondimensional).

Continuity:

$$\frac{\partial (r\, u_r)}{\partial r} + r\, \frac{\partial u_z}{\partial z} = 0$$

Radial momentum:

$$u_r u_{r,r} + u_z u_{r,z} - \frac{u_\theta^2}{r} + p_{\mathrm{coeff}}\, p_{,r} - \frac{1}{Re}\left[u_{r,rr} + \frac{u_{r,r}}{r} - \frac{u_r}{r^2} + u_{r,zz}\right] = 0$$

Axial momentum:

$$u_r u_{z,r} + u_z u_{z,z} + p_{\mathrm{coeff}}\, p_{,z} - \frac{1}{Re}\left[u_{z,rr} + \frac{u_{z,r}}{r} + u_{z,zz}\right] - \frac{Gr}{Re^2}\,\theta = 0$$

Governing equations (2/2): swirl and energy

Swirl (angular momentum):

$$u_r u_{\theta,r} + u_z u_{\theta,z} + \frac{u_r u_\theta}{r} - \frac{1}{Re}\left[u_{\theta,rr} + \frac{u_{\theta,r}}{r} - \frac{u_\theta}{r^2} + u_{\theta,zz}\right] = 0$$

Energy:

$$u_r \theta_{,r} + u_z \theta_{,z} - \frac{1}{Pr\,Re}\left[\theta_{,rr} + \frac{\theta_{,r}}{r} + \theta_{,zz}\right] = 0$$

$$\theta = \frac{T - T_{\min}}{T_{\max} - T_{\min}}, \qquad Re = 15, \quad Pr = 28.463, \quad Gr = 8000, \quad p_{\text{coeff}} = 134{,}128$$

Physics loss and hard constraints

- Derivatives via `torch.autograd.grad`, `create_graph=True` (second order).
- Five residuals (continuity, r, z, swirl, energy), balanced by EMA weighting, $\beta = 0.95$.
- Staged schedule: data only until epoch 50, then physics weight ramped $0 \to 0.05$.
- Gradient clipping at L2 norm 1.0.
- **Hard constraints:** $u_r = r \cdot \mathrm{raw}_0$, $u_\theta = r \cdot \mathrm{raw}_2$, so both vanish on the axis exactly.

Current limitation

The PQC output is detached inside the physics loss, so PDE gradients do not reach the quantum angles. The PQC is currently shaped by the data loss only.

Fix in progress

Parameter-shift or adjoint gradients for the residual path.

From QPINN to QPINO: learning across operating conditions

QPINN (today)

One model per operating condition:
$(r, z) \mapsto (u_r, u_z, u_\theta, p, T)$.

QPINO (next)

The process parameters become inputs:

$$(r, z, s) \;\mapsto\; (u_r, u_z, u_\theta, p, T), \quad s \in \{T_{\mathrm{hot}},\, \omega_c,\, \omega_s,\, v_{\mathrm{pull}}\}$$

One trained surrogate serves the whole sweep: the basis for parametric exploration and digital twins.

[IMAGE PLACEHOLDER]
`images/qpino-concept.png`
QPINO: operating parameters s enter the encoder alongside (r, z); one model, many conditions

Benchmark 1: silicon CZ melt (running)

Property	Value
Reference	Axisymmetric CFD
Domain	$r, z \in [0, 0.3]$ m
Points	18,108
Outputs	u_r, u_z, u_θ, p, T
T range	1685–1750 K
Split	17,203 / 905

[IMAGE PLACEHOLDER]
images/training-data-fields.png
Training data fields (normalised): u_r, u_z, u_θ, p, T over (r, z)

Scope

Steady, laminar, single-phase reference cases, not fully resolved industrial melt flow.

Next physics

Oxygen concentration transport (species equation) and the evolving melt–crystal interface.

Benchmark 2: LH_2 cryogenic boil-off (defined)

Physics to capture

- Wall heat leak $\rightarrow$ natural convection boundary layer and thermal stratification.
- Turbulent convection (RANS closure in the reference CFD).
- Liquid–vapour interface with evaporation (VOF + phase-change mass transfer).
- Outputs: velocity, temperature, phase fraction, boil-off rate, ullage pressure.

Status

Benchmark and outputs defined; reference CFD data not yet generated. **Planned**

[IMAGE PLACEHOLDER]

`images/lh2-boiloff-cfd.png`

LH_2 tank CFD: temperature stratification and interface (reference, to be generated)

Evaluation protocol: what counts as evidence

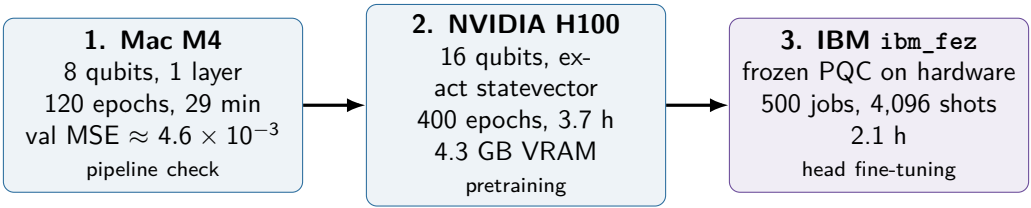
- **Ablations:** (a) classical-only, matched parameter budget; (b) random-feature layer in place of the PQC.
- **Baselines:** classical PINN and PINO with matched optimiser, sampling, seeds, budget.
- **Generalisation:** case-wise holdout of a whole operating condition.
- **Reporting:** per-field error and unweighted loss components.
- **Cost:** wall time to target error and inference latency.

Sweep	Range	Held out
T_{hot}	1730–1785 K	1780 K
ω_c (crucible)	-10 to -1 rpm	-4 rpm
ω_s (crystal)	4–20 rpm	7 rpm

Why this matters

A random spatial split measures interpolation. A held-out operating condition measures what a digital twin actually needs.

Compute pipeline: laptop → GPU → quantum hardware



Note on simulation

A 16-qubit circuit is a 2^{16}-amplitude statevector: exactly simulable on one GPU. Hardware runs here test noise robustness and deployment, not quantum advantage.

Result 1: fast convergence in the data-driven phase **Shown**

Epoch	Train loss	Val MSE	Val T
1	1.16×10^{-2}	4.83×10^{-3}	1.37×10^{-3}
10	5.92×10^{-5}	3.11×10^{-4}	1.47×10^{-4}
20	$\mathbf{5.26 \times 10^{-5}}$	$\mathbf{1.58 \times 10^{-4}}$	$\mathbf{5.07 \times 10^{-5}}$
30	2.96×10^{-5}	1.75×10^{-4}	6.15×10^{-5}
50	2.07×10^{-5}	2.13×10^{-4}	2.91×10^{-5}

Reading

Validation MSE drops 30× in 20 epochs. Best checkpoint: $\mathbf{1.58 \times 10^{-4}}$ (random spatial split).

[IMAGE PLACEHOLDER]

`images/training-curve-v2.png`

Train / validation loss vs epoch (H100 run, 400 epochs); physics ramp start marked at epoch 50

Result 2: all five coupled fields captured **Shown**

Field	Val MSE	
u_r	1.93×10^{-6}	excellent
u_z	5.78×10^{-5}	very good
u_θ	1.44×10^{-6}	excellent
p	4.62×10^{-5}	good
T	5.07×10^{-5}	good

Best checkpoint (epoch 20), normalised fields, in distribution.

[IMAGE PLACEHOLDER]
`images/parity-best-checkpoint.png`
Prediction vs ground truth per field, best checkpoint (parity plots)

[IMAGE PLACEHOLDER]
`images/field-error-maps.png`
Absolute error maps over (r, z) for u_r, u_z, u_θ, p, T

Result 3: naive physics weighting destabilises training **Shown**

Field	Best (ep. 20)	Final (ep. 400)	Factor
u_r	1.93×10^{-6}	1.79×10^{-4}	93×
u_z	5.78×10^{-5}	8.83×10^{-3}	153×
u_θ	1.44×10^{-6}	5.06×10^{-5}	35×
p	4.62×10^{-5}	0.100	**2,165**×
T	5.07×10^{-5}	0.129	**2,544**×

```
[0280] loss=1.04e+09 data=5.06e-02 phys=2.16e+10
[0290] loss=1.18e+14 data=4.97e-02 phys=2.15e+15
[0300] loss=5.57e-02 data=4.85e-02 phys=9.62e-02
```

Diagnosis

Data loss stays near 0.05; the blow-up is entirely in the physics residual. Root cause: the p_{coeff} scale mismatch (Bottleneck 3).

Fixes under test **In progress**

Physics weight 0.001–0.005; per-field pressure residual scaling; re-check pressure nondimensionalisation.

Result 4: end-to-end run on real quantum hardware **Shown**

Metric	Value
Backend	`ibm_fez` (156 qubits)
Jobs completed / failed	**500 / 0**
Shots per job	4,096
Total shots	**2,048,000**
Session	125 min, 4.0 jobs/min
Trainable scope	head only (19,205)
Calibration points	8

Session stopped by plan time at ≈62 of 100 planned epochs; all results stored under saved job IDs.

[IMAGE PLACEHOLDER]
`images/calibration-coverage.png`
Calibration subset coverage: 8 points across the (r, z) domain

Hardware noise and why the head is fine-tuned

Qubit	$\langle Z \rangle$	σ	σ_{shot}
Q_0	$+0.342$	0.012	0.015
Q_1	$+0.765$	0.014	0.010
Q_5	-0.856	0.009	0.008
Q_{13}	-0.908	0.007	0.007

$\sigma_{\text{shot}} = \sqrt{(1 - \langle Z \rangle^2)/4096}$; reported spread is close to shot noise.

Adaptation strategy

Freeze the PQC; retrain only the classical head on hardware $\langle Z_i \rangle$.

[IMAGE PLACEHOLDER]

`images/initial-predictions-before-finetune.png`

Initial predictions vs ground truth, before fine-tuning (near-constant outputs)

Observation

Before fine-tuning, predictions on hardware features are nearly flat across all five fields: the head must be re-adapted. Completing this run is the next hardware step.

Inside the trained circuit

[IMAGE PLACEHOLDER]
`images/quantum-rot-params.png`
Rot gate angles $(\varphi, \vartheta, \omega)$ per qubit, layers 0 and 1

- Angles span the full $[0, 2\pi]$ range (mean 3.49, std 1.79 rad).
- No qubit stuck at a degenerate value.

[IMAGE PLACEHOLDER]
`images/quantum-layernorm.png`
Quantum LayerNorm scale γ and bias β per qubit

- In the longer v1 run (650 epochs), LayerNorm weights qubits unequally ($\gamma \approx 0.3$–1.2): the head learns to rely on some quantum features more than others.
- At the v2 best checkpoint (epoch 20) γ is still near 1.

The circuit is active and differentiated. Whether it adds accuracy is what the ablation will tell us.

Benchmark program: convergence, generalisation, cost

[IMAGE PLACEHOLDER]
`images/bench-convergence.png`
Val error vs epoch: Hybrid-QPINN vs classical PINN vs random-feature control

[IMAGE PLACEHOLDER]
`images/bench-generalisation.png`
Per-field NRMSE on held-out operating conditions: QPINO vs PINO

[IMAGE PLACEHOLDER]
`images/bench-cost.png`
Training cost and inference latency: hybrid vs classical

Convergence **In progress**	Generalisation **Planned**	Cost **Planned**
Epochs and wall time to reach a target per-field error, matched budgets.	Case-wise holdout on the three CZ sweeps.	GPU-hours to target error; CPU inference time per full field.

These three panels are where the hybrid-vs-classical claims will be decided

Our early data-only results converge quickly, but a matched classical baseline is required before we claim an advantage.

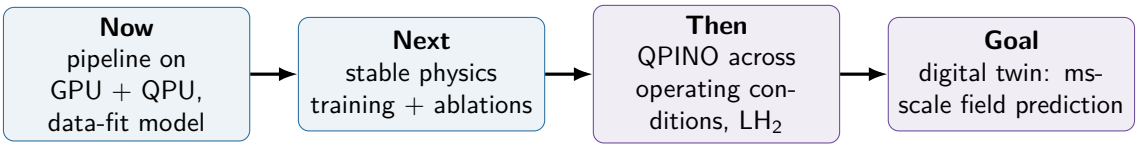
Where the evidence stands today

Claim	Status	Evidence
Hybrid pipeline runs on GPU and real QPU	**Shown**	500/500 IBM jobs, no failures
Five coupled CZ fields fitted in distribution	**Shown**	Val MSE 1.58×10^{-4}
Fast early convergence	**Shown**	$30\times$ drop in 20 epochs
Faster convergence than classical PINN/PINO	**In progress**	Matched baseline running
PQC adds accuracy over classical features	**In progress**	Ablation + random-feature control
Sharper generalisation across conditions	**Planned**	Case-wise holdout, QPINO
Reduced computational cost	**Planned**	GPU-hours to target error
Concentration and evolving interfaces	**Planned**	Species eq., interface tracking
LH_2 boil-off	**Planned**	Benchmark defined

Next experiments

Priority	Experiment	Compute
Critical	Classical-only baseline, matched parameters	$1\times$H100, $\sim$4 h
Critical	Random-feature control in place of the PQC	$1\times$H100, $\sim$4 h
Critical	Retrain with physics weight 0.001–0.005	$1\times$H100, $\sim$4 h
Critical	Retrieve the 500 IBM job results	minimal
High	Physics gradients through the PQC (parameter shift)	architecture change
High	Case-wise holdout on three CZ sweeps (QPINO)	3 runs
High	Complete head fine-tuning on IBM hardware	$\sim$500 jobs
Medium	LH_2 boil-off reference CFD + surrogate	CFD + H100

Pathway to real-time digital twins



Data engine

Agentic CFD (AutoFOAM) generates verified training cases from one prompt.

Surrogate

Hybrid-QPINO trained once, queried across the full process window.

Validator

Physics and benchmark checks before any prediction is trusted.

Key takeaways

1. **Framework:** Fourier features + data re-uploading + a shallow 16-qubit PQC as a trainable feature map, inside a classical physics-informed network.
2. **Demonstrated:** the hybrid model fits all five coupled CZ fields (val MSE 1.58×10^{-4}) and runs end to end on IBM hardware (500 jobs, 2.05 M shots, no failures).
3. **Learned:** physics weighting must respect field scales; a pressure coefficient of $\sim 10^5$ can wreck training.
4. **Being tested:** faster convergence, better generalisation and lower cost than classical PINN/PINO, via matched ablations.
5. **Destination:** QPINO surrogates for crystal growth and LH_2 storage as the core of real-time digital twins.

References

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Thank you

Questions?

Hybrid Quantum AI for accelerating multiphysics simulation